

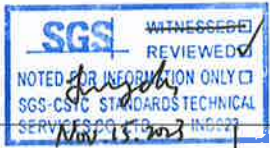
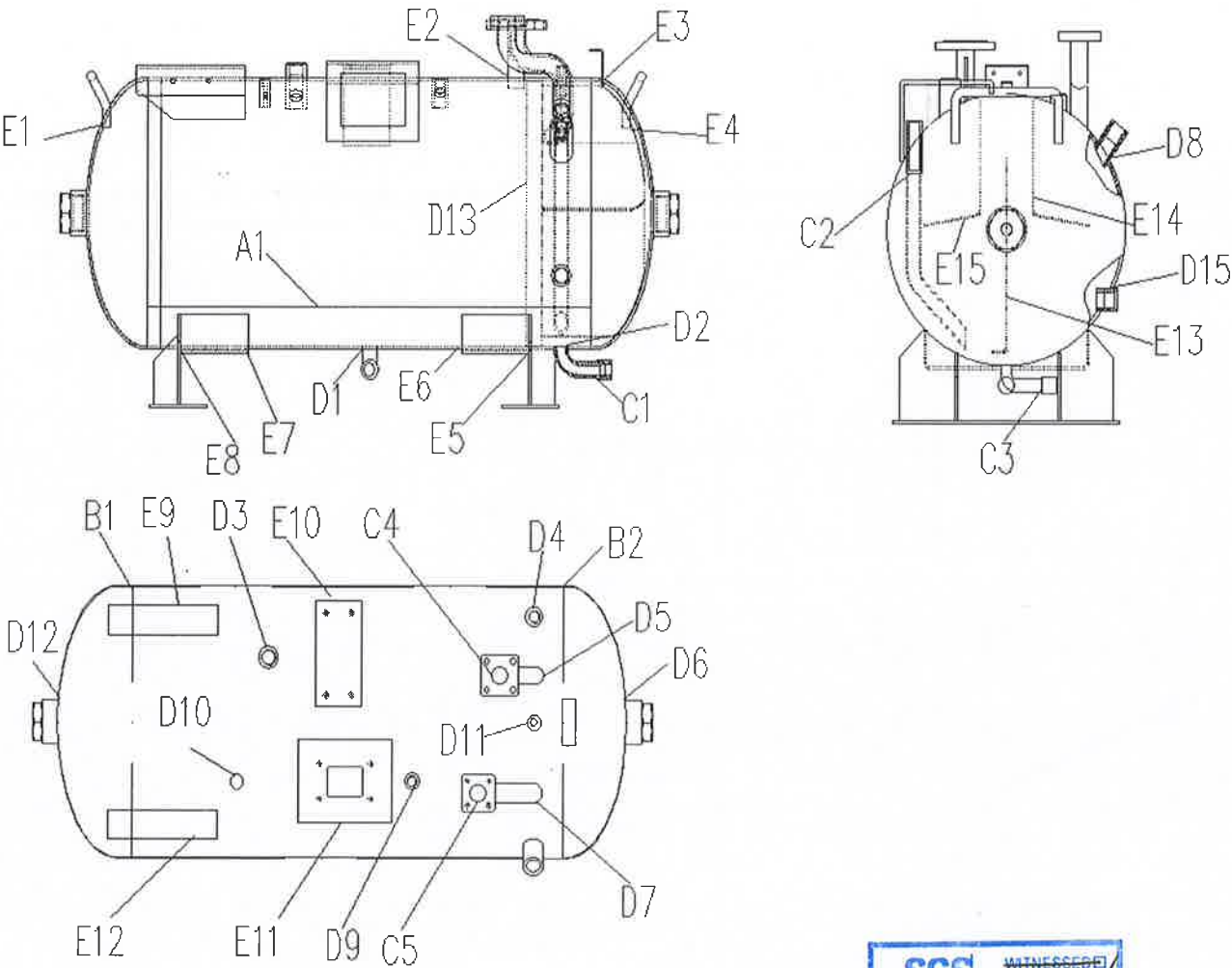
WTAH

5678OINT IDENTIFICATION CARD (JIC)
焊缝识别卡

Exhibit 7-7 (A) , Edition 3, Revision 0
Doc. No.: JIC-P-2362 (A) Page 1 of 1

工作令号 Job No.	P-2362	产品名称 Product Name	油气分离器 OIL/GAS SEPARATOR	产品图号 Drawing No./ Rev. No	20980145-083 Rev.3
-----------------	--------	----------------------	----------------------------	------------------------------	-----------------------

设备焊缝识别简图 Joint Identification Sketch of Item



			2023.11.10		2023.11.10
修改号 Rev. No.	修改说明 Rev. Description	焊接工程师编制 Prepared By Welding Engineer	日期 Date	技术经理审核 Reviewed By Technical Manager	日期 Date

WTAH

JOINT IDENTIFICATION CARD (JIC)

焊缝识别卡

Exhibit 7-7 (B), Edition 3, Revision 0

Doc. No. JIC-P-2362 (B) Page 1 of 1

焊缝编号 Joint No.	母材 Base Material	WPS 号 WPS No.	修订号 Rev. No.	焊工位置资格 Welder'S Position Required	备注 Remarks
A1	SA/GB713 Q345R	WPS-TK23-19	0	PA 平	121
B1/B2	SA/GB713 Q345R	WPS-TK23-19	0	PA 平	121
C1~C3	SA-105 MM & SA-106 Gr. B	WPS-TK22-08	0	PA 平/PB 横	141
C4/C5	SA-105 MM & SA/GB713 Q345R	WPS-TK23-17	0	PA 平/PB 横	141
D13	SA/GB713 Q345R	WPS-TK23-18	0	PA 平/PB 横	135
D1/D2 D5/D7	SA/GB713 Q345R & SA-106 Gr. B	WPS-TK23-15	0	PA 平/PB 横	Root(打底): 141 Capping(盖面): 135
D3/D4/D6 D8~D12	SA/GB713 Q345R & SA-105 M	WPS-TK23-15	0	PA 平/PB 横	Root(打底): 141 Capping(盖面): 135
E1/E4	SA/GB713 Q345R & SA-105 M	WPS-TK23-17	0	PA 平/PB 横	141
E2/E3 E5~E15	SA/GB713 Q345R & Q235B	WPS-TK23-18	0	PA 平/PB 横	135



焊工 Welder

工钢印 Welder's Stamp No.	焊工姓名 Welder's Name	焊接方法 Welding Process	焊缝编号 Joint No
CH07	祁治国	121	A1/B1/B2
CH09	郑诗松	141	C1~C5/ D1~D12/E1/E4
CH10	张宝山	135	D1~D13/E2/E3/E5~E15

For all welds if no defect ,the tack weld can be incorporated into welds if unacceptable removed.

对所有的焊缝点焊, 若无缺陷则可以熔入焊缝, 若有缺陷则应当消除

焊接工程师编制 Prepared By Welding Technician		日期 Date	2023.11.10
技术经理审核 Reviewed By Technical Manager		日期 Date	2023.11.10



TANKER

Welding Procedure Specification (WPS)

焊接工艺规程

Welding Procedure
specification 工艺规程编号

WPS-TK23-19 Rev.0

WPQR No

WPQR-TK22-18

工艺评定号

Manufacturer

Well-Tanker(Anhui) M&E
Equipment Co., Ltd.

制造商

Mode of metal transfer

N/A

金属过渡形式

Throat thickness(mm)

N/A

焊喉厚度

Joint Type and weld Type

Butt Joint and Butt Weld

接头形式和焊缝形式

对接接头/对接焊缝

Weld Preparation Details (Sketch) 焊缝准备详图

Method of Preparation and cleaning

坡口制备及清理

Parent Material Designation

母材规格

Material thickness(mm)

材料厚度

Outside pipe Diameter(mm)

接管外径

Deposited Weld Metal

熔敷金属

Welding position

焊接位置

Machine and Grinding

机加工和打磨

SA/GB713 Q345R, Group: 1.2

G235B, Group: 1.1

3~7.8mm

Fixed > 500: Rot. > 150mm

Max. 7.8mm

EN ISO 6947-PA

Joint Design 接头设计	Welding sequences 焊接顺序
See detailed fabrication drawing 见制造详图	See detailed fabrication drawing 见制造详图

Welding Details 焊接

Run 焊层	Welding process 焊接方法	Filler Size 焊丝直径	Current 电流 A	Voltage 电压 V	Type of Current/polarity 电流极性	Wire feed/Travel speed 送丝速度	Run out length/ travel speed 焊接速度 mm/s	Arc energy Heat input 线能量/热输入 KJ/mm
1	121	2.5mm	300-340	28-32	DC EP	20mm	2.0-2.5	≤4.95

Filler material designation and make

填充金属类型及牌号

NB/T47018.4-H10Mn2

Any Special Baking of Drying

烘干

250 °C 2h (Flux)

Designation Gas/ Flux 气体/焊剂:

NB/T47018.4-F5A2-H10Mn2

- Shielding 保护气

None 无

- Backing 背保护

None 无

Gas Flow Rate 气体流量:

- Shielding 保护气

None 无

- Backing 背保护

None 无

Tungsten electrode type / size

钨极类型/尺寸

None 无

Details of back gouging / backing

背面清根/垫磨

Carbon grouping, grouping

碳弧气刨, 打磨

Preheat temperature 预热温度

≥5 °C

Interpass temperature 层间温度

≤ 300 °C

Post heating 后处理

None 无

Post weld heat treatment and/or Ageing

(time, temp, method, heating & cooling

rates) 焊后热处理 (保温时间, 温度,

方法, 升温降温速率)

None 无

Other information, e.g.

其他信息

None 无

Weaving (max width of run)

摆动 (最大宽度)

NA

Oscillation: amplitude

frequency, dwell time,

Pulse welding details

摆动: 振幅, 频率, 停留时间

NA

Distance contact tube/work

piece 管嘴道工件距离

15~25mm

Pulse Welding details 脉冲焊

接

NA

Plasma weld details 等离子焊

NA

Torch angle 焊枪角度

NA

Pre heat maintenance

temperature 预热保持温度

NA

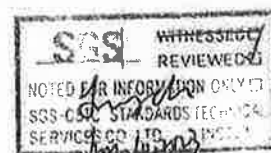
Manufacturer 制造商
(name, signature, date)Well-Tanker (Anhui) M&E Equipment Co., Ltd.
安徽威尔泰克机电设备有限公司

编制 Prepared by: 孙

批准 Approved by: 孙

日期 Date: 2023.6.10

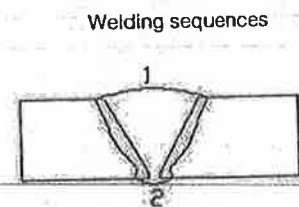
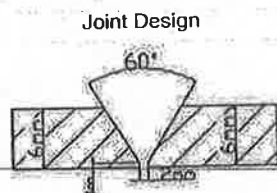
日期 Date: 2023.6.10



	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Record of Weld Test		

Location	Hefei, China	Examiner or Test body	HPi Verification Services (Ireland) Ltd
Procedure Ref No	pWPS-TK22-18/Rev.0	Method of Preparation	Machine and Grinding
Manufacturer's WPQR No	WPQR-TK22-18	Material specification	GB/T 713 Q345R
Manufacturer	Well-Tanker (Anhui) M&E Equipment Co., Ltd.	Material thickness	6 mm
Welder's Name	Qi Zhiguo /CH07	*Outside pipe dia	NA
Welding Process	EN ISO 4063- 121	Welding position	EN ISO 6947-PA
Joint Type	Butt joint		

Weld Preparation Details (Sketch)



Welding Details

Run	Process	Filler Size mm	Current A	Voltage V	Type of Current	Wire-feed/Travel speed mm/s	Heat input KJ/mm
1-end	121	2.5	330	31	DC EP	2.0	5.11

Filler material type & name

NB/T47018.4 H10Mn2
(Tianjin Golden Bridge
Welding Materials Group
Co., Ltd.)

Other information: None

Baking or drying

250°C, 2hr (Flux)

Weaving (max width of run)

N/A

Gas-flux / rate - Shielding

NB/T47018.4
F5A2-H10Mn2

Oscillation: amplitude
frequency, dwell time,

N/A

Backing

N/A

Tungsten electrode type / size

N/A

Pulse welding details

N/A

Details of back gouging / backing

Grinding

Stand-off distance

N/A

Preheat temperature

Min 5°C

Plasma weld details

N/A

Interpass temperature

Max 300 °C

Torch angle

N/A

Post heating Post weld heat treatment
(time, temp, method, heating & cooling rates)

N/A

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code/testing standard(s) indicated above.

Manufacturer Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Examiner

M Convento CEng

Signature

Signature

Name

Jiang Mingxin

Name of Examining Body

HPi Verification Services
(Ireland) Ltd

Reference Number

iP1014-034- WPQR-TK22-18

Date

22 August 2022

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Welding Procedure Qualification Record (WPQR)		

Code/Testing standard EN 15614-1: 2017

Manufacturer

Address

Ref No: HPiVS/iP1014-034- WPQR-TK22-18
 Well-Tanker (Anhui) M&E Equipment Co., Ltd.
 Middle Section of Baihua Avenue, Wushan Town,
 Changfeng County, Hefei City, Anhui Province,
 People's Republic of China

Manufacturer's WPQR No

Date of Welding

WPQR-TK22-18

7 August 2022

Range of Qualification:

Welding Process/es

Joint and weld type

Parent Material group

121 Fully Mechanized Welding
 BW/ FW
 ISO/TR 15608:2013 Group1-1,
 (min. specified yield strength $\leq 345\text{MPa}$)

Parent Material Thickness (mm)

Weld metal thickness (mm)

Throat thickness

Outside pipe diameter (mm)

Filler material designation/make

Shielding gas/ flux designation

Type of welding current and polarity

Welding positions

Preheat temperature (min)

Interpass Temperature (max)

Post weld heat treatment

Other information: Refer to Sec.8 of the ISO 15614-1

$3 \leq t \leq 7.8$ (Single -run)

Max.12

No restriction

Fixed > 500 ; Rot, > 150

NB/T47018.4 H10Mn2

NB/T47018.4 F5A2-H10Mn2

DCEP

All except PG, PJ and J-L045

Min 5°C

$\leq 300^{\circ}\text{C}$

None

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard(s) indicated above.

Examiner

Signature

M. Convento CEng



Name of Examining Body

Reference Number

Date

HPi Verification Services (Ireland) Ltd
 Clonross, Dunshaughlin, Co.Meath
 Ireland

iP1014-068- WPQR-TK22-18

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	Welding Procedure Test Results		

Manufacturer's WPQR No	WPQR-TK22-18	Examiner or Test body	HPI Verification Services (Ireland) Ltd
Visual	Acceptable	Reference No	IP1014-034- WPQR-TK22-18
Penetrant / Magnetic Particle	Acceptable	Radiography/Ultrasonie	Acceptable

Tensile Tests – EN ISO 4136

Temperature: 18 °C

Type / No	Re N/mm ²	Rm N/mm ²	A% on	Z%	Fracture	Remarks
TK-P-22-18-2	/	580	/	/	Weld Metal	Ductile
TK-P-22-18-3	/	580	/	/	Weld Metal	Ductile

Bend Tests – EN ISO 5173

Type / No	Bend Angle	Elongation	Results
Face bend/ TK-P-22-18-4, -5	180	N/A	No defect
Root bend/ TK-P-22-18-6, -7	180	N/A	No defect

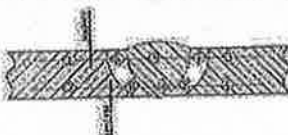
Macroscopic Examination – EN ISO 17639

Acceptable

Impact Test – EN ISO 9016

Notch Location / Direction	Temp C	Values			Average	Remarks
		1	2	3		
Weld Metal	0	84J	72J	91J	82J	Specimen Size: 5x10x55mm
HAZ	0	92J	73J	67J	77J	Specimen Size: 5x10x55mm

Hardness Test – EN ISO 9015-1

Load Type: HV10	Location of Measurements (Sketch)
Parent Metal 147~160	
HAZ 156~160	
Weld Metal 172~187	

Other Tests:

Tests carried out in accordance with the requirements of
Laboratory report reference no

Test results

Test carried out in the presence of

Examiner or Examining Body

Name, date & Signature

ISO 15614-1:2017

TK22-18-VT18, LPR-TK22-18-PT18, RER-TK22-18-RT18,
WPS-WTAH-2218

Acceptable

Zhang Peng / 037

HPI Verification Services (Ireland) Ltd.

M Convento CEng, 29 August 2022





TANKER

Welding Procedure Specification (WPS)

接工艺规程

Welding Procedure
specification 工艺规程编号

WPS-TK23-08 Rev.0

WPQR No

WPQR-TK22-09

工艺评定号

Manufacturer

Well-Tanker(Anhui) M&E
Equipment Co., Ltd.

制造商

Mode of metal transfer

N/A

金属过渡形式

Throat thickness(mm)

N/A

焊喉厚度

Joint Type and weld Type

Butt Joint and Butt Weld

接头形式和焊缝形式

对接接头/对接焊缝

Weld Preparation Details (Sketch) 焊缝准备详图

Method of Preparation and cleaning

坡口制备及清理

Parent Material Designation

母材规格

Material thickness(mm)

材料厚度

Outside pipe Diameter(mm)

接管外径

Deposited Weld Metal

熔敷金属

Welding position

焊接位置

Machine and Grinding

机加工和打磨

SA-105 M Group: 11.1

SA-106 B Group: 11.1

 $3 \leq t \leq 11.08 \text{mm}$ $\geq 16.7 \text{mm}$

Max.11.08mm

EN ISO 6947-H-L045

Joint Design 接头设计		Welding sequences 焊接顺序	
	FW: See detailed fabrication drawing 见制造详图	BW: See detailed fabrication drawing 见制造详图	FW: See detailed fabrication drawing 见制造详图

Welding Details 焊接

Run 焊层	Welding process 焊接方法	Filler Size 焊丝直径	Current 电流 A	Voltage 电压 V	Type of Current/polarity 电流极性	Wire feed/Travel speed 送丝速度	Run out length/ travel speed 焊接速度 mm/s	Arc energy Heat input 线能量/热输入 KJ/mm
1(打底)	141	2.0mm	110-140	14-18	DC EN	N/A	1.0-2.0	≤ 0.880
		2.5mm	120-150	14-19			1.1-2.1	
2 ~ end	141	2.0mm	130-170	15-20	DC EN	N/A	1.0-2.5	≤ 1.824
		2.5mm	150-180	16-21			1.1-2.6	
		3.2mm	170-200	18-23			1.2-2.8	

Filler material designation and make
填充金属类型及牌号

AWS A5.18 ER70S-6

Any Special Baking of Drying
烘干

N/A

Other information, e.g.
其他信息

None 无

Designation Gas/ Flux 气体/焊:

- Shielding 保护气
- Backing 背保护

Ar (99.99%)

Weaving (max width of run)
摆动 (最大宽度)

Max. 6mm

Gas Flow Rate 气体流量:

- Shielding 保护气
- Backing 背保护

N/A

Oscillation: amplitude
frequency, dwell time,
Pulse welding details
摆动: 振幅, 频率, 停留时间

N/A

Tungsten electrode type / size
钨极类型/尺寸W-Ce20, Dia. 1.6/2.0/2.5mm
钨钨极, 直径 1.6/2.0/2.5mmDetails of back gouging / backing
背面清根/垫板

None 无

Distance contact tube/work
piece 管嘴道工件距离

N/A

Preheat temperature 预热温度

 $\geq 5 \text{ } ^\circ\text{C}$

Pulse Welding details 脉冲焊接

N/A

Interpass temperature 层间温度

 $\leq 300 \text{ } ^\circ\text{C}$

Plasma weld details 等离子焊

N/A

Post heating 后处理

N/A

Torch angle 焊枪角度

N/A

Post weld heat treatment and/or Ageing
(time, temp, method, heating & cooling
rates) 焊后热处理 (保温时间, 温度, 方
法, 升温降温速率)

N/A

Pre heat maintenance
temperature 预热保持温度

N/A

Manufacturer 制造商

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

(name, signature, date)

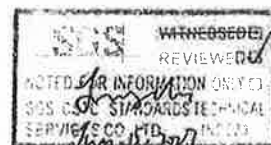
安徽威尔泰克机电设备有限公司

编制 Prepared by: 

日期 Date: 2023.6.10

批准 Approved by: 

日期 Date: 2023.6.10

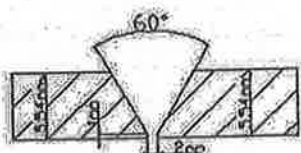


	REFERENCE:	PRE - F - 23	ISSUE:	02
	TITLE:	BS EN ISO 15614-1 Record of Weld Test		

Location	Hefei, China	Examiner or Test body	HPiVS
Procedure Ref No	pWPS-TK22-09/Rev.0	Method of Preparation	Machine and Grinding
Manufacturer's WPQR No	WPQR-TK22-09	Material specification	SA-106 Gr.B
Manufacturer	Well-Tanker (Anhui) M&E Equipment Co., Ltd.	Material thickness	5.54 mm
Welder's Name	Zheng Shisong /CH09	Outside pipe dia	33.4 mm
Welding Process	EN ISO 4063- 141	Welding position	EN ISO 6947-H-L045
Joint Type	Butt joint		

Weld Preparation Details (Sketch)

Joint Design



Welding sequences



Welding Details

Run	Process	Filler-Size mm	Current A	Voltage V	Type of Current	Wire-feed/Travel speed mm/s	Heat input KJ/mm
1	141	2.0	110	16	DC EN	1.2	0.880
2 ~ end	141	2.0	160	19	DC EN	1.0	1.824

Filler material type & name

AWS A5.18 ER70S-6
(Tianjin Golden Bridge
Welding Materials Group
Co., Ltd.)

Other information: None

Baking or drying

N/A

Weaving (max width of run)

Max. 8.0mm

Gas-flux / rate – Shielding

Ar (99.99%)12-16 L/min

Oscillation: amplitude

N/A

Backing

N/A

frequency, dwell time,

Tungsten electrode type / size

W-Ce20, Dia.2.0mm

Pulse welding details

N/A

Details of back gouging / backing

Grinding

Stand-off distance

N/A

Preheat temperature

Min 5°C

Plasma weld details

N/A

Interpass temperature

Max 300 °C

Torch angle

N/A

Post

Post weld heat treatment

N/A

heating

(time, temp, method, heating & cooling rates)

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code/testing standard(s) indicated above.

Manufacturer Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Examiner

M Convento CEng

Signature

Signature

Name

Jiang Mingxing

Name of Examining Body

HPi Verification Services
(Ireland) Ltd

Reference Number

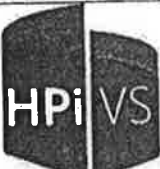
iP1014-068- WPQR-TK22-09

Date

6 April 2022

Date

8 April 2022

	REFERENCE:	PRE - F - 23	ISSUE:	02
	TITLE:	BS EN ISO 15614-1 Welding Procedure Qualification Record (WPQR)		

Code/Testing standard EN 15614-1: 2017

Manufacturer

Address

Manufacturer's WPQR No

Date of Welding

Ref No: iP1014-068- WPQR-TK22-09

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Middle Section of Baihua Avenue, Wushan Town,
Changfeng County, Hefei City, Anhui Province,
People's Republic of China

WPQR-TK22-09

11 March 2022

Range of Qualification:

Welding Process/es

Joint and weld type

Parent Material group

Parent Material Thickness (mm)

Weld metal thickness (mm)

Throat thickness

Outside pipe diameter (mm)

Filler material designation/make

Shielding gas-/ flux designation

Type of welding current and polarity

Welding positions

Preheat temperature (min)

Interpass Temperature (max)

Post weld heat treatment

Other information: Refer to Sec.8 of the ISO 15614-1

141 Manual Welding

BW/ FW

ISO/TR 15608:2013 Group 11.1-11.1,11.1-1,
1-1(min. specified yield strength $\leq 260\text{MPa}$)

$3 \leq t \leq 11.08$ (Multi -run)

Max.11.08

No restriction

≥ 16.7

AWS A5.18 ER70S-6

Ar (99.99%)

DCEN

All except PG, PJ and J-L045

Min 5°C

≤ 300 °C

None

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard(s) indicated above.

Examiner

Signature

M Convento



Name of Examining Body

HPi Verification Services (Ireland) Ltd
Clonross, Dunshaughlin, Co. Meath
Ireland

Reference Number

iP1014-068- WPQR-TK22-09

Date

8 Apr 2022

	REFERENCE:	PRE - F - 23	ISSUE:	02
	TITLE:	Welding Procedure Test Results		

Manufacturer's WPQR No WPQR-TK22-09 Examiner or Test body HPI Verification Services
 Visual Acceptable Reference No iP1014-068- WPQR-TK22-09
 Penetrant / Magnetic Particle Acceptable Radiography/Ultrasonic Acceptable

Tensile Tests – EN ISO 4136

Temperature: 18 °C

Type / No	Re N/mm ²	Rm N/mm ²	A% on	Z%	Fracture	Remarks
TK-P-22-9-2	/	445	/	/	Base Metal	Ductile
TK-P-22-9-3	/	475	/	/	Base Metal	Ductile

Bend Tests – EN ISO 5173

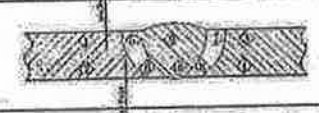
Type / No	Bend Angle	Elongation	Results
Face bend/ TK-P-22-9-4, -5	180	N/A	No defect
Root bend/ TK-P-22-9-6, -7	180	N/A	No defect (except 2x1mm imperfection on TK-P-22-9-6)

Macroscopic Examination – EN ISO 17639 Acceptable

Impact Test – EN ISO 9016

Notch Location / Direction	Temp C	Values			Average	Remarks
		1	2	3		
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

Hardness Test – EN ISO 9015-1

Load Type: HV10	Location of Measurements (Sketch)
Parent Metal 148~160	
HAZ 154~167	
Weld Metal 193~227	

Other Tests:

Tests carried out in accordance with the requirements of
 Laboratory report reference no

ISO 15614-1:2017

Test results

TK22-09-VT09, LPR-TK22-09-PT09, RER-TK22-09-RT09,
 WPS-WTAH-2209
 Acceptable

Test carried out in the presence of

Zhang Peng / ID 039

Examiner or Examining Body

HPI Verification Services (Ireland) Ltd..

Name, date & Signature

M Convento CEng, 8 April 2020





TANKER

Welding Procedure Specification (WPS)

接工艺规程

Welding Procedure
specification 工艺规程编号
WPQR No
工艺评定号

WPS-TK23-17 Rev.0

WPQR-TK22-16

Manufacturer
制造商

Well-Tanker(Anhui) M&E
Equipment Co., Ltd.

Mode of metal transfer
金属过渡形式

N/A

Throat thickness(mm)
焊喉厚度

N/A

Joint Type and weld Type
接头形式和焊缝形式

FW
角接

Weld Preparation Details (Sketch) 焊缝准备详图

Method of Preparation and cleaning
坡口制备及清理
Parent Material Designation
母材规格

Machine and Grinding
机加工和打磨
SA-516 Gr70, Group: 11.1
SA-105M, Group: 11.1
SA/GB713 Q345R Group: 1.2

Material thickness(mm)
材料厚度

t1: 3~12mm
t2: 3~28mm

Outside pipe Diameter(mm)
接管外径

N/A

Deposited Weld Metal
熔敷金属

N/A

Welding position
焊接位置

EN ISO 6947-PB

Joint Design 接头设计	Welding sequences 焊接顺序
See detailed fabrication drawing 见制造详图	See detailed fabrication drawing 见制造详图

Welding Details 焊接

Run 焊层	Welding process 焊接方法	Filler Size 焊丝直径	Current 电流 A	Voltage 电压 V	Type of Current/polarity 电流极性	Wire feed/Travel speed 送丝速度	Run out length/ travel speed 焊接速度 mm/s	Arc energy Heat input 线能量/热输入 KJ/mm
1(打底)	141	2.0mm	110-140	14-18	DC EN	N/A	1.0-2.0	≤0.768
		2.5mm	120-150	14-19			1.1~2.1	
2 ~ end	141	2.0mm	130-170	15-20	DC EN	N/A	1.0-2.5	≤1.067
		2.5mm	150-180	16-21			1.1-2.6	
		3.2mm	170-200	18-23			1.2-2.8	

Filler material designation and make
填充金属类型及牌号

AWS A5.18 ER70S-6

Any Special Baking of Drying
烘干

N/A

Other information, e.g.
其他信息

None 无

Designation Gas/ Flux 气体/焊:

- Shielding 保护气
- Backing 背保护

Ar (99.99%)

Weaving (max width of run)
摆动 (最大宽度)

Max. 6mm

Gas Flow Rate 气体流量:

- Shielding 保护气
- Backing 背保护

N/A

Oscillation: amplitude
frequency, dwell time,
Pulse welding details
摆动: 振幅, 频率, 停留时间

N/A

Tungsten electrode type / size
钨极类型/尺寸

W-Ce20, Dia. 1.6/2.0/2.5mm
钨钨极, 直径 1.6/2.0/2.5mm

Details of back gouging / backing
背面清根/垫板

None 无

Distance contact tube/work
piece 管嘴道工件距离

N/A

Preheat temperature 预热温度

≥5 °C

Pulse Welding details 脉冲焊
接

N/A

Interpass temperature 层间温度

≤ 300 °C

Plasma weld details 等离子焊

N/A

Post heating 后处理

N/A

Torch angle 焊枪角度

N/A

Post weld heat treatment and/or Ageing
(time, temp, method, heating & cooling
rates) 焊后热处理 (保温时间, 温度,
方法, 升温降温速率)

N/A

Pre heat maintenance
temperature 预热保持温度

N/A

Manufacturer 制造商
(name, signature, date)

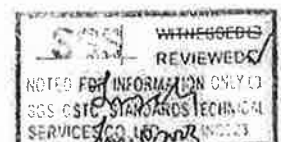
Well-Tanker (Anhui) M&E Equipment Co., Ltd.
安徽威尔泰克机电设备有限公司

编制 Prepared by:

日期 Date: 2023.6.10

批准 Approved by:

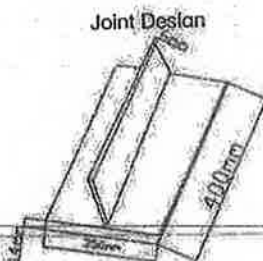
日期 Date: 2023.6.10



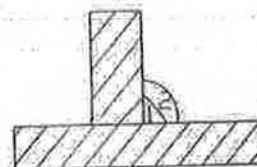
	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Record of Weld Test		

Location	Hefei, China	Examiner or Test body	HPi Verification Services (Ireland) Ltd
Procedure Ref No	pWPS-TK22-16/Rev.0	Method of Preparation	Machine and Grinding
Manufacturer's WPQR No	WPQR-TK22-16	Material specification	SA-516 Gr.70/GB/T 713 Q345R
Manufacturer	Well-Tanker (Anhui) M&E Equipment Co., Ltd.	Material thickness	SA-516 Gr.70t1=6mm, GB/T 713 Q345R t2=14mm
Welder's Name	Deng Zhigang /CH04	Outside pipe dia	NA
Welding Process	EN ISO 4063- 141	Welding position	PB
Joint Type	T Joint (FW ml)		

Weld Preparation Details (Sketch)



Welding sequences



Welding Details

Run	Process	Filler Size mm	Current A	Voltage V	Type of Current	Wire feed/Travel speed mm/s	Heat input KJ/mm
1	141	2.0	120	16	DC EP	1.5	0.768
2	141	2.0	170	20	DC EP	1.8	1.067

Filler material type & name

AWS A5.18 ER70S-6
(Tianjin Golden Bridge
Welding Materials Group
Co., Ltd.)

Other information:

Baking or drying

N/A

Weaving (max width of run)

Max. 6.0mm

Gas flux / rate – Shielding
Backing

Ar (99.99%) 10-20 L/min
N/A

Oscillation: amplitude
frequency, dwell time,

N/A

Tungsten electrode type / size

W-Ce20, Dia.2.0mm

Pulse welding details

N/A

Details of back gouging / backing

Grinding

Stand-off distance

N/A

Preheat temperature

Min 5°C

Plasma weld details

N/A

Interpass temperature

Max 300 °C

Torch angle

N/A

Post heating Post weld heat treatment
(time, temp, method, heating & cooling rates)

N/A

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code/testing standard(s) indicated above.

Manufacturer Well-Tanker (Anhui) M&E Equipment Co., Ltd. Examiner

Signature

Signature

Name Jiang Mingxin

Name of Examining Body HPi Verification Services (Ireland) Ltd

Reference Number

IP1014-034- WPQR-TK22-16

Date 22 August 2022

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Welding Procedure Qualification Record (WPQR)		

Code/Testing standard EN 15614-1: 2017

Manufacturer

Address

Manufacturer's WPQR No

Date of Welding

Ref No: HPiVS/iP1014-034- WPQR-TK22-16

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Middle Section of Baihua Avenue, Wushan Town,
Changfeng County, Hefei City, Anhui Province,
People's Republic of China

WPQR-TK22-16

7 August 2022

Range of Qualification:

Welding Process/es

Joint and weld type

Parent Material group

141 Manual Welding

FW

ISO/TR 15608:2013 Group 11.1-1,1-1(min.
specified yield strength $\leq 345\text{MPa}$ for Group 1; min.
specified yield strength $\leq 260\text{MPa}$ for Group 11.1)

$3 \leq t_1 \leq 12, 3 \leq t_2 \leq 28$

N/A

No restriction

N/A

AWS A5.18 ER70S-6

Ar (99.99%)

DCEN

All except PG, PJ and J-L045

Min 5°C

$\leq 300^\circ\text{C}$

None

Parent Material Thickness (mm)

Weld metal thickness (mm)

Throat thickness

Outside pipe diameter (mm)

Filler material designation/make

Shielding gas / flux designation

Type of welding current and polarity

Welding positions

Preheat temperature (min)

Interpass Temperature (max)

Post weld heat treatment

Other information: Refer to Sec.8 of the ISO 15614-1

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard(s) indicated above.

Examiner

Signature

M Convento CEng



Name of Examining Body

HPi Verification Services (Ireland) Ltd
Clonross, Dunshaughlin, Co.Meath
Ireland

Reference Number

iP1014-034- WPQR-TK22-16

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	Welding Procedure Test Results		

Manufacturer's WPQR No	WPQR-TK22-16	Examiner or Test body	HPi Verification Services (Ireland) Ltd
Visual	Acceptable	Reference No	iP1014-034- WPQR-TK22-16
Penetrant / Magnetic Particle	Acceptable	Radiography/Ultrasonic	N/A

Tensile Tests – EN ISO 4136

Temperature: 18 °C

Type / No	Re N/mm ²	Rm N/mm ²	A% on	Z%	Fracture	Remarks
/	/	/	/	/	/	/
/	/	/	/	/	/	/

Bend Tests – EN ISO 5173

Type / No	Bend Angle	Elongation	Results
/	/	/	/
/	/	/	/

Macroscopic Examination – EN ISO 17639

Acceptable

Impact Test – EN ISO 9016

Notch Location / Direction	Temp C	Values			Average	Remarks
		1	2	3		
/	/	/	/	/	/	/
/	/	/	/	/	/	/

Hardness Test – EN ISO 9015-1

Load Type: HV10	Location of Measurements (Sketch)
Parent Metal 183~210	
HAZ 195~202	
Weld Metal 212~216	

Other Tests:

Tests carried out in accordance with the requirements of
Laboratory report reference no
Test results

ISO15614-1:2017

TK22-16-VT16, LPR-TK22-16-PT16, WPS-WTAH-2216
Acceptable

Test carried out in the presence of

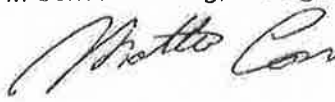
Zhang Peng / 037

Examiner or Examining Body

HPi Verification Services (Ireland) Ltd

Name, date & Signature

M Convento CEng, 29 August 2022



Welding Procedure Specification (WPS)

焊接工艺规程

Welding Procedure specification 工艺规程编号

WPS-TK23-18 Rev.0

WPQR No

WPQR-TK22-17

工艺评定号

Manufacturer

Well-Tanker(Anhui) M&E

制造商

Equipment Co., Ltd.

Mode of metal transfer

Globular/ Spray 熔滴/ 喷射

金属过渡形式

Throat thickness(mm)

N/A

焊喉厚度

Joint Type and weld Type

FW

接头形式和焊缝形式

角接

Weld Preparation Details (Sketch) 焊缝准备详图

Method of Preparation and cleaning

坡口制备及清理

Parent Material Designation

母材规格

Material thickness(mm)

材料厚度

Outside pipe Diameter(mm)

接管外径

Deposited Weld Metal

熔敷金属

Welding position

焊接位置

Machine and Grinding

机加工和打磨

Q235B, Group: 1.1

SA/GB713Q345R Group: 1.2

t1: 3~12mm

t1: 3~56mm

N/A

N/A

NA

NA

EN ISO 6947-PB

Joint Design 接头设计	Welding sequences 焊接顺序
See detailed fabrication drawing 见制造详图	See detailed fabrication drawing 见制造详图

Welding Details 焊接

Run 焊层	Welding process 焊接方法	Filler Size 焊丝直径	Current 电流 A	Voltage 电压 V	Type of Current/polarit y 电流极性	Wire feed/Travel speed 送丝速度	Run out length/ travel speed 焊接速度 mm/s	Arc energy Heat input 线能量/热输入 KJ/mm
1 (打底)	135	1.0mm	130-150	20-30	DC EP	N/A	1.5-2.5	≤1.188
		1.2mm	150-170	20-30	DC EP		1.8-2.7	
2 ~ end	135	1.0mm	170-190	25-33	DC EP	N/A	1.5-3.0	≤1.613
		1.2mm	190-210	25-33	DC EP		2.0-3.2	

Filler material designation and make

填充金属类型及牌号

NB/T47018.3-ER50-6

Any Special Baking of Drying

烘干

NA

Designation Gas/ Flux 气体/焊:

- Shielding 保护气

75%Ar/25%CO₂

- Backing 背保护

None 无

Gas Flow Rate: 气体流量:

- Shielding 保护气

15~20L/Min

- Backing 背保护

None 无

Tungsten electrode type / size

钨极类型/尺寸

NA

Details of back gouging / backing

背面清根/垫板

None 无

Preheat temperature 预热温度

≥5 °C

Interpass temperature 层间温度

≤ 300 °C

Post heating 后处理

None 无

Post weld heat treatment and/or Ageing
(time, temp, method, heating & cooling
rates) 焊后热处理 (保温时间, 温度,

方法, 升温降温速率)

None 无

Other information, e.g.

其他信息

None 无

Weaving (max width of

run)

摆动 (最大宽度)

Max.3.0mm

Oscillation: amplitude

frequency, dwell time,

Pulse welding details

摆动: 振幅, 频率, 停留时

间

NA

Distance contact

tube/work piece 管嘴道工

件距离

NA

Pulse Welding details 脉冲

焊接

NA

Plasma weld details 等离

子焊

NA

Torch angle 焊枪角度

NA

Pre heat maintenance

temperature 预热保持温度

NA

Manufacturer 制造商

(name, signature, date)

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

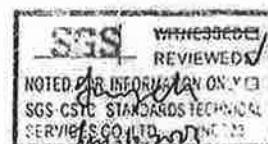
安徽威尔泰克机电设备有限公司

编制 Prepared by:

批准 Approved by:

日期 Date: 2023.6.10

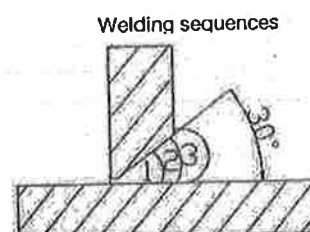
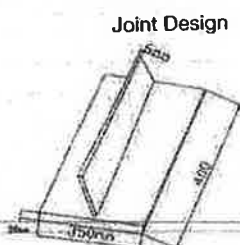
日期 Date: 2023.6.10



	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Record of Weld Test		

Location	Hefei, China	Examiner or Test body	HPi Verification Services (Ireland) Ltd
Procedure Ref No	pWPS-TK22-17/Rev.0	Method of Preparation	Machine and Grinding
Manufacturer's WPQR No	WPQR-TK22-17	Material specification	GB/T 713 Q345R
Manufacturer	Well-Tanker (Anhui) M&E Equipment Co., Ltd.	Material thickness	t1=6mm, t2=28mm
Welder's Name	Zhang Baoshan /CH10	Outside pipe dia	NA
Welding Process	EN ISO 4063- 135	Welding position	PB
Joint Type	T joint (BW+FW)		

Weld Preparation Details (Sketch)



Welding Details

Run	Process	Filler Size mm	Current A	Voltage V	Type of Current	Wire feed/Travel speed mm/s	Heat input KJ/mm
1	135	1.0	135	22	DC EP	2.0	1.188
2-cap	135	1.0	180	28	DC EP	2.5	1.613

Filler material type & name

NB/T 47018.3 ER50-6
(Tianjin Golden Bridge
Welding Materials Group
Co., Ltd.)

Other Information:

Baking or drying

N/A

Weaving (max width of run)

Max. 3.0mm

Gas flux / rate - Shielding

75%Ar +25% CO₂ 15-20
L/min

Oscillation: amplitude
frequency, dwell time,

N/A

Backing

N/A

Pulse welding details

N/A

Tungsten electrode type / size

N/A

Stand-off distance

N/A

Details of back gouging / backing

Grinding

Plasma weld details

N/A

Preheat temperature

Min 5°C

Torch angle

N/A

Interpass temperature

Max 300 °C

N/A

Post heating Post weld heat treatment
(time, temp, method, heating & cooling rates)

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code/testing standard(s) indicated above.

Manufacturer Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Examiner

M Convento CEng

Signature

Signature



Name Jiang Mingxin

Name of Examining Body

HPi Verification Services
(Ireland) Ltd

Reference Number

iP1014-034- WPQR-TK22-17

Date 22 August 2022

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Welding Procedure Qualification Record (WPQR)		

Code/Testing standard EN 15614-1: 2017

Manufacturer

Address

Manufacturer's WPQR No

Date of Welding

Ref No: HPiVS/P1014-034- WPQR-TK22-17

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Middle Section of Baihua Avenue, Wushan Town,
Changfeng County, Hefei City, Anhui Province,
People's Republic of China

WPQR-TK22-17

7 August 2022

Range of Qualification:

Welding Process/es

135 Partly Mechanized Welding (Mode of metal transfer: Spray, Plus & Globular)

Joint and weld type

T Joint (BW+FW)

Parent Material group

ISO/TR-15608:2013-Group1-1(min. specified yield strength $\leq 345\text{MPa}$)

Parent Material Thickness (mm)

$3 \leq t_1 \leq 12, 3 \leq t_2 \leq 56$

Weld metal thickness (mm)

Max. 12

Throat thickness

No restriction

Outside pipe diameter (mm)

N/A

Filler material designation/make

NB/T47018.3 ER50-6

Shielding gas / flux designation

75%Ar +25% CO₂

Type of welding current and polarity

DCEP

Welding positions

All except PG, PJ and J-L045

Preheat temperature (min)

Min 5°C

Interpass Temperature (max)

$\leq 300^\circ\text{C}$

Post weld heat treatment

None

Other information: Refer to Sec.8 of the ISO 15614-1

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard(s) indicated above.

Examiner

M Convento CEng

Signature



Name of Examining Body

HPi Verification Services (Ireland) Ltd
Clonross, Dunshaughlin, Co.Meath
Ireland

Reference Number

iP1014-034- WPQR-TK22-17

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	Welding Procedure Test Results		

Manufacturer's WPQR	WPQR-TK22-17	Examiner or Test body	HPi Verification Services (Ireland) Ltd
No		Reference No	IP1014-034- WPQR-TK22-17
Visual	Acceptable	Radiography/Ultrasonic	Acceptable
Penetrant / Magnetic Particle	Acceptable		

Tensile Tests – EN ISO 4136					Temperature: 18 °C	
Type / No	Re N/mm ²	Rm N/mm ²	A% on	Z%	Fracture	Remarks
/	/	/	/	/	/	/
/	/	/	/	/	/	/

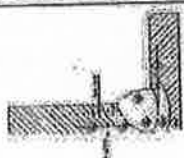
Bend Tests – EN ISO 5173			
Type / No	Bend Angle	Elongation	Results
/	/	/	/

Macroscopic Examination – EN ISO 17639 Acceptable

Impact Test – EN ISO 9016

Notch Location / Direction	Temp C	Values			Average	Remarks
		1	2	3		
/	/	/	/	/	/	/
/	/	/	/	/	/	/

Hardness Test – EN ISO 9015-1

Load Type: HV10	Location of Measurements (Sketch)
Parent Metal 183–210	
HAZ 195–202	
Weld Metal 212–216	

Other Tests:

Tests carried out in accordance with the requirements of
Laboratory report reference no

Test results

Test carried out in the presence of

Examiner or Examining Body

Name, date & Signature

ISO15614-1:2017

TK22-17-VT17, LPR-TK22-17-PT17, RER-TK22-17-RT17,
WPS-WTAH-2217

Acceptable

Zhang Peng / 037

HPi Verification Services (Ireland) Ltd.

M Convento CEng, 29 August 2017





TANKER

Welding Procedure Specification (WPS)

焊接工艺规程

Welding Procedure
specification 工艺规程编号

WPQR No

工艺评定号

Manufacturer

制造商

Mode of metal transfer

金属过渡形式

Throat thickness(mm)

焊喉厚度

Joint Type and weld Type

接头形式和焊缝形式

WPS-TK23-15 Rev.0

WPQR-TK22-14

WPQR-TK22-15

Well-Tanker(Anhui) M&E

Equipment Co., Ltd.

135: Golbular/ Spray

135: 熔滴/ 喷射

NA

30° Branch connection/ Butt

Weld

支管连接/对接焊缝

Method of Preparation and cleaning

坡口制备及清理

Parent Material Designation

母材规格

Material thickness(mm)

材料厚度

Outside pipe Diameter(mm)

接管外径

Deposited Weld Metal

熔敷金属

Welding position

焊接位置

Machine and/or Grinding

机加工和/或打磨

SA-105M Gr. B, Group: 11.1

SA/GB713 Q345R Group: 1.2

Plate 3~12mm

Pipe 3-10.16mm

Main pipe: ≥109.5mm,

Branch pipe: ≥24.15mm

Max.12mm

EN ISO 6947-PA+PB

Weld Preparation Details (Sketch) 焊缝准备详图

Joint Design 接头设计	Welding sequences 焊接顺序
See detailed fabrication drawing 见制造详图	See detailed fabrication drawing 见制造详图

Welding Details 焊接

Run 焊层	Welding process 焊接方法	Filler Size 焊丝直径	Current 电流 A	Voltage 电压 V	Type of Current/polarity 电流极性	Wire feed/Travel Speed mm/s 送丝速度	Run out length/ travel speed 焊接速度 mm/s	Arc energy Heat input 线能量/热输入 KJ/mm
1(打底)	141	2.0mm	140-170	14-18	DC EN	NA	1-2	≤0.900
		2.5mm	150-180	15-19			1.1-2.1	
2~end	135	1.0mm	170-190	25-33	DC EP	20	1.5-3.0	≤2.418
		1.2mm	190-210	27-35		24	1.7-3.2	

Filler material designation and make
填充金属类型及牌号AWS A5.18 ER70S-6(141)
NB/T47018.3-ER50-6(135)Any Special Baking or Drying
烘干

NA

Designation Gas/ Flux 气体/焊:

- Shielding 保护气

141:Ar(99.99%)
135:75%Ar/25%CO2

- Backing 背保护

None 无

Gas Flow Rate 气体流量:

- Shielding 保护气

141:10~20L/Min
135: 15~20L/Min

- Backing 背保护

None 无

Tungsten electrode type / size
钨极类型/尺寸141:W-Ce20, Dia. 1.6/2.0/2.5mm
141: 铈钨极, 直径 1.6/2.0/2.5mmDetails of back gouging / backing
背面清根/垫板

None 无

Preheat temperature 预热温度

≥5 °C

Interpass temperature 层间温度

≤ 300 °C

Post heating 后处理

None 无

Post weld heat treatment and/or
Ageing(time, temp, method, heating &
cooling rates) 焊后热处理 (保温时
间, 温度, 方法, 升温降温速率)

None 无

Manufacturer 制造商

(name, signature, date)

Well-Tanker (Anhui) M&E Equipment Co., Ltd.
安徽威尔泰克机电设备有限公司

Other information, e.g.

其他信息

None 无

Weaving (max width of run)
摆动 (最大宽度)141:Max.6.0mm
135: Max.3.0mmOscillation: amplitude
frequency, dwell time,

Pulse welding details

摆动: 振幅, 频率, 停留时间

NA

Distance contact tube/work
piece 管嘴道工件距离

NA

Pulse Welding details 脉冲焊接

NA

Plasma weld details 等离子焊

NA

Torch angle 焊枪角度

NA

Pre heat maintenance

temperature 预热保持温度

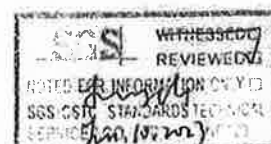
NA

编制 Prepared by:

批准 Approved by:

日期 Date: 2023.6.10

日期 Date: 2023.6.10

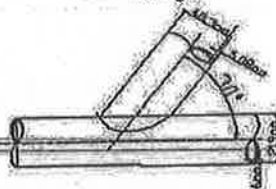


	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Record of Weld Test		

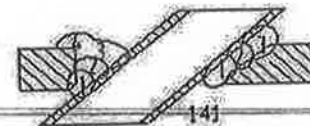
Location	Hefei, China	Examiner or Test body	HPI Verification Services (Ireland) Ltd
Procedure Ref No	pWPS-TK22-14/Rev.0	Method of Preparation	Machine and Grinding
Manufacturer's WPQR No	WPQR-TK22-14	Material specification	GB/T 713 Q345R/SA-106 Gr.B
Manufacturer	Well-Tanker (Anhui) M&E Equipment Co., Ltd.	Material thickness	Main pipe t1:6 mm/ Branch pipe t2:5.08 mm (SA-106 Gr.B)
Welder's Name	Zheng Shisong /CH09	Outside pipe dia	Main pipe :219mm, Branch pipe: 48.3 mm (SA-106 Gr.B)
Welding Process	EN ISO 4063- 141	Welding position	EN ISO 6947-PA+PB
Joint Type	Branch Connection		

Weld Preparation Details (Sketch)

Joint Design



Welding sequences



Welding Details

Run	Process	Filler Size mm	Current A	Voltage V	Type of Current	Wire-feed/Travel speed mm/s	Heat Input KJ/mm
1	141	2.0	120	15	DC EN	1.2	0.900
2~ end	141	2.0	160	20	DC EN	1.1	1.745

Filler material type & name

AWS A5.18 ER70S-6,
(Tianjin Golden Bridge
Welding Materials Group
Co., Ltd.)

Other information: None

Baking or drying

N/A

Weaving (max width of run)

Max. 6.0mm

Gas-flux / rate - Shielding

Ar (99.99%)10-20 L/min

Oscillation: amplitude

N/A

Backing

N/A

frequency, dwell time,

Tungsten electrode type / size

W-Ce20, Dia.2.0mm

Pulse welding details

N/A

Details of back gouging / backing

Grinding

Stand-off distance

N/A

Preheat temperature

Min 5°C

Plasma weld details

N/A

Interpass temperature

Max 300 °C

Torch angle

N/A

Post

Post weld heat treatment

N/A

heating

(time, temp, method, heating & cooling rates)

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code/testing standard(s) indicated above.

Manufacturer Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Examiner

M Convento CEng

Signature

Signature

Name

Jiang Mingxin

Name of Examining Body

HPI Verification Services
(Ireland) Ltd

Reference Number

ip1014-034- WPQR-TK22-14

Date

22 August 2022

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Welding Procedure Qualification Record (WPQR)		

Code/Testing standard EN 15614-1: 2017

Manufacturer

Address

Ref No: HPIVS/iP1014-034- WPQR-TK22-14

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Middle Section of Baihua Avenue, Wushan Town,
Changfeng County, Hefei City, Anhui Province,
People's Republic of China

WPQR-TK22-14

7 August 2022

Manufacturer's WPQR No

Date of Welding

Range of Qualification:

Welding Process/es

Joint and weld type

141 Manual Welding

Butt welds from one or both sides and fillet welds
on plate or pipe (branch angle $\geq 30^\circ$)

Parent Material group

ISO/TR 15608:2013 Group 11.1-1,1-1(min.

specified yield strength $\leq 345\text{MPa}$ for Group1; min.
specified yield strength $\leq 260\text{MPa}$ for Group11.1)

Parent Material Thickness (mm)

3 to 12 (Multi -run)

Weld metal thickness (mm)

Max.12

Throat thickness

No restriction

Outside pipe diameter (mm)

≥ 24.15

Filler material designation/make

AWS A5.18 ER70S-6

Shielding gas/ flux designation

Ar (99.99%)

Type of welding current and polarity

DCEN

Welding positions

All except PG, PJ and J-L045

Preheat temperature (min)

Min 5°C

Interpass Temperature (max)

$\leq 300^\circ\text{C}$

Post weld heat treatment

None

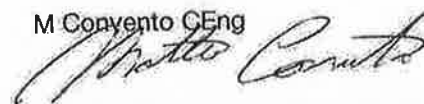
Other information: Refer to Sec.8 of the ISO 15614-1

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard(s) indicated above.

Examiner

M Convento CEng

Signature



Name of Examining Body

HPi Verification Services (Ireland) Ltd
Clonross, Dunshaughlin, Co.Meath
Ireland

Reference Number

iP1014-034- WPQR-TK22-14

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	Welding Procedure Test Results		

Manufacturer's WPQR No	WPQR-TK22-14	Examiner or Test body	HPi Verification Services (Ireland) Ltd
Visual	Acceptable	Reference No	iP1014-034- WPQR-TK22-14
Penetrant / Magnetic Particle	Acceptable	Radiography/Ultrasonic	Acceptable

Tensile Tests – EN ISO 4136
Temperature: 18 °C

Type / No	Re N/mm ²	Rm N/mm ²	A% on	Z%	Fracture	Remarks
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

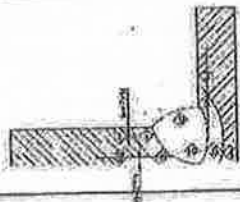
Bend Tests – EN ISO 5173

Type / No	Bend Angle	Elongation	Results
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A

Macroscopic Examination – EN ISO 17639
Acceptable
Impact Test – EN ISO 9016

Notch Location / Direction	Temp C	Values			Average	Remarks
		1	2	3		
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

Hardness Test – EN ISO 9015-1

Load Type: HV10	Location of Measurements (Sketch)
Parent Metal 150~168	
HAZ 165~175	
Weld Metal 174~177	

Other Tests:

Tests carried out in accordance with the requirements of Laboratory report reference no

Test results

Test carried out in the presence of

Examiner or Examining Body
Name, date & Signature

ISO 15614-1:2017

TK22-14-VT14, LPR-TK22-14-PT14, RER-TK22-14-RT14,
WPS-WTAH-2214

Acceptable

Zhang Peng / 037

HPi Verification Services (Ireland) Ltd
M Convento CEng, 29 August 2022

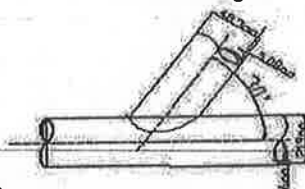



	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Record of Weld Test		

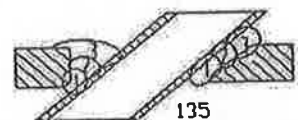
Location	Hefei, China	Examiner or Test body	HPI Verification Services (Ireland) Ltd
Procedure Ref No	pWPS-TK22-15/Rev.0	Method of Preparation	Machine and Grinding
Manufacturer's WPQR No	WPQR-TK22-15	Material specification	GB/T 713 Q345R/SA-106 Gr.B
Manufacturer	Well-Tanker (Anhui) M&E Equipment Co., Ltd.	Material thickness	Main pipe t1:6 mm/ Branch pipe t2:5.08 mm (SA-106 Gr.B)
Welder's Name	Zhang Baoshan /CH10	Outside pipe dia	Main pipe :219mm, Branch pipe 48.3 mm (SA-106 Gr.B)
Welding Process	EN ISO 4063- 135(Mode of metal transfer: Globular)	Welding position	EN ISO 6947-PA+PB
Joint Type	Branch Connection		

Weld Preparation Details (Sketch)

Joint Design



Welding sequences



Welding Details

Run	Process	Filler Size mm	Current A	Voltage V	Type of Current	Wire-feed/Travel speed mm/s	Heat input KJ/mm
1	135	1.0	130	21	DC EP	1.7	1.285
2~ end	135	1.0	170	32	DC EP	1.8	2.418

Filler material type & name

NB/T 47018.3 ER50-6
(Tianjin Golden Bridge
Welding Materials Group
Co., Ltd.)

Other information: None

Baking or drying

N/A

Weaving (max width of run)

Max. 3.0mm

Gas-flux / rate - Shielding

75%Ar +25% CO₂ 15-20
L/min

Oscillation: amplitude,
frequency, dwell time,

N/A

Backing

N/A

Pulse welding details

N/A

Tungsten electrode type / size

N/A

Stand-off distance

N/A

Details of back gouging / backing

Grinding

Plasma weld details

N/A

Preheat temperature

Min 5°C

Torch angle

N/A

Interpass temperature

Max 300 °C

N/A

Post heating

Post weld heat treatment

(time, temp, method, heating & cooling rates)

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code/testing standard(s) indicated above.

Manufacturer Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Examiner

M Convento CEng

Signature

Signature

Name

Jiang Mingxin

Name of Examining Body

HPI Verification Services
(Ireland) Ltd

Reference Number

iP1014-034- WPQR-TK22-15

Date

22 August 2022

Date

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	BS EN ISO 15614-1 Welding Procedure Qualification Record (WPQR)		

Code/Testing standard EN 15614-1: 2017

Manufacturer

Address

Manufacturer's WPQR No

Date of Welding

Ref No: HPiVS/iP1014-034- WPQR-TK22-15

Well-Tanker (Anhui) M&E Equipment Co., Ltd.

Middle Section of Baihua Avenue, Wushan Town,
Changfeng County, Hefei City, Anhui Province,
People's Republic of China

WPQR-TK22-15

7 August 2022

Range of Qualification:

Welding Process/es

Joint and weld type

Parent Material group

Parent Material Thickness (mm)

Weld metal thickness (mm)

Throat thickness

Outside pipe diameter (mm)

Filler material designation/make

Shielding gas/ flux designation

Type of welding current and polarity

Welding positions

Preheat temperature (min)

Interpass Temperature (max)

Post weld heat treatment

Other information: Refer to Sec.8 of the ISO 15614-1

135 Partly Mechanized Welding (Mode of metal transfer: Spray, Plus & Globular)

Butt welds from one or both sides and filler welds on plate or pipe (branch angle $\geq 30^\circ$)

ISO/TR 15608:2013 Group 11.1-1.1-1(min. specified yield strength $\leq 345\text{MPa}$ for Group1; min. specified yield strength $\leq 260\text{MPa}$ for Group11.1)

3 to 12 (Multi -run)

Max.12

No restriction

≥ 24.15

NB/T47018.3 ER50-6

75%Ar +25% CO₂

DCEP

All except PG, PJ and J-L045

Min 5°C

$\leq 300^\circ\text{C}$

None

I certify that test welds were prepared, welded and tested satisfactorily in accordance with the requirements of the code / testing standard(s) indicated above.

Examiner

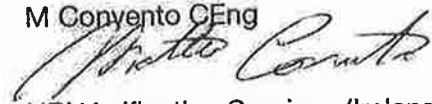
Signature

Name of Examining Body

Reference Number

Date

M Convento CEng



HPi Verification Services (Ireland) Ltd

Clonross, Dunshaughlin, Co.Meath

Ireland

iP1014-034- WPQR-TK22-15

29 August 2022

	REFERENCE:	PRE - F - 23	ISSUE:	03
	TITLE:	Welding Procedure Test Results		

Manufacturer's WPQR No	WPQR-TK22-15	Examiner or Test body	HPI Verification Services (Ireland) Ltd
Visual	Acceptable	Reference No	IP1014-034- WPQR-TK22-15
Penetrant / Magnetic Particle	Acceptable	Radiography/Ultrasonic	Acceptable

Tensile Tests – EN ISO 4136
Temperature: 18 °C

Type / No	Re N/mm ²	Rm N/mm ²	A% on	Z%	Fracture	Remarks
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

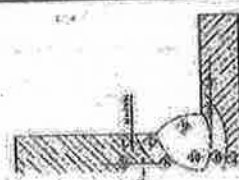
Bend Tests – EN ISO 5173

Type / No	Bend Angle	Elongation	Results
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A

Macroscopic Examination – EN ISO 17639
Acceptable
Impact Test – EN ISO 9016

Notch Location / Direction	Temp C	Values			Average	Remarks
		1	2	3		
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

Hardness Test – EN ISO 9015-1

Load Type: HV10	Location of Measurements (Sketch)
Parent Metal 142~169	
HAZ 170~180	
Weld Metal 174~182	

Other Tests:

Tests carried out in accordance with the requirements of Laboratory report reference no

ISO15614-1:2017

TK22-15-VT15, LPR-TK22-15-PT15, RER-TK22-15-RT15, WPS-WTAH-2215

Test results

Acceptable

Test carried out in the presence of

Zhang Peng / 037

Examiner or Examining Body

HPI Verification Services (Ireland) Ltd

Name, date & Signature

M Convento CEng, 29 August 2022



	REFERENCE:	PRE - F - 24	ISSUE:	01
	TITLE:	ISO 9606-1 Welder Qualification Test Certificate		

Designation: 141 T/P Branch FM1 S/B t6 D48.3 PB ss nb WPS No.: pWPS-TK22-06 Rev.0 Welder's Name: Zheng Shisong/CH09 Identification: 340121198204234012 Method of Identification: ID Card Date and place of birth: 23 April 1982, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: ISO 9606-1:2017	Examiner or Test Body Reference No. iP1014-034-TK22-06-ZS 
Job Knowledge: Acceptable ☐ Not tested ☒	

	Weld test details	Range of approval
Welding process(es)	141	141/142/143/145
Transfer mode	N/A	N/A
Product type (Plate or pipe)	Plate, Pipe	Plate/Pipe, Pipe
Type of weld	Branch (Set-through)	Branch
Parent Material group(s)	11.1 to 11.1	1~11
Filler Material group(s)	FM1 (S)	FM1, FM2 (S, M)
Shielding gases	99.99%Ar	99.99%Ar
Auxiliaries (eg backing gases)	Without	With or Without
Type of current and polarity	DCEN	DCEN
Deposited thickness (mm)	6	3 to 12
Outside pipe diameter (mm)	48.3	≥25
Welding position	PA+PB	PA, PB, PA+PB
Welding details	ss, nb	ss nb, ss mb, bs, ss gb, ss fb

Supplementary fillet weld test is available on attached sheet and or document No. NA

Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiographic testing	☐	☒
Fracture test	☐	☒
Bend test	☐	☒
Notch tensile test	☐	☒
Macroscopic exam	☒	☐
other	☐	☒

Examiner or test body: HPiVS Ireland

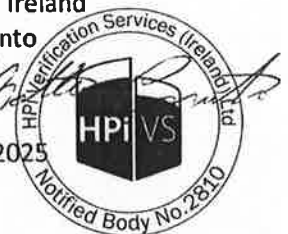
Date of issue: 11 April 2022

Location: Clonross, Ireland

Name of Examiner: M Convento

Signature: 

Validity of approval until 05 April 2025



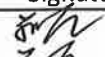
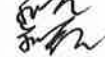
Revalidation in accordance to

9.3 a) 3 YEARS

Prolongation for approval by examiner or test body for the following 2 years

Date	Signature	Position or title

Prolongation for approval by employer/co-ordinator the following 6 months

Date	Signature	Position or title
2022.9.28		welding engineer
2023.3.20		welding engineer
2023.9.15		welding engineer

	REFERENCE:	PRE - F - 24	ISSUE:	01
	TITLE:	ISO 9606-1 Welder Qualification Test Certificate		

Designation: 141 P FW FM1 S t6 PB ml WPS No.: pWPS-TK22-12 Rev.0 Welder's Name: Zheng Shisong/CH09 Identification: 340121198204234012 Method of Identification: ID Card Date and place of birth: 23 April 1982, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: ISO 9606-1:2017	Examiner or Test Body Reference No. iP1014-034-TK22-12-ZS 
Job Knowledge: Acceptable ☐ Not tested ☒	

	Weld test details	Range of approval
Welding process(es)	141	141/142/143/145
Transfer mode	N/A	N/A
Product type (Plate or pipe)	Plate	Plate / Pipe
Type of weld	FW	FW
Parent Material group(s)	11.1 to 11.1	1~11
Filler Material group(s)	FM1 (S)	FM1, FM2 (S, M)
Shielding gases	99.99%Ar	99.99%Ar
Auxiliaries (eg backing gases)	Without	With or Without
Type of current and polarity	DCEN	DCEN
Material thickness (mm)	6	≥3
Outside pipe diameter (mm)	--	≥75 rotating, or ≥500 for fixed pipes
Welding position	PB	PA, PB
Welding details	ml	sl, ml

Supplementary fillet weld test is available on attached sheet and or document No. NA

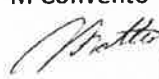
Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiographic testing	☐	☒
Fracture test	☐	☒
Bend test	☐	☒
Notch tensile test	☐	☒
Macroscopic exam	☒	☐
other	☐	☒

Examiner or test body: HPi Verification Services (Ireland)

Date of issue: 24 April 2022

Location: Co. Meath, Ireland

Name of Examiner: M Convento

Signature: 

Validity of approval until 18 Apr. 2025



Revalidation in accordance to

9.3 a) 3 YEARS

Prolongation for approval by examiner or test body for the following 2 years

Date	Signature	Position or title

Prolongation for approval by employer/co-ordinator the following 6 months

Date	Signature	Position or title
2022-9-15		welding officer
2023-3-20		welding engineer
2023-9-15		welding engineer

	REFERENCE:	PRE - F - 24	ISSUE:	01
	TITLE:	ISO 9606-1 Welder Qualification Test Certificate		

Designation: 135 P FW FM1 S t6 PB ml WPS No.: pWPS-TK22-11 Rev.0 Welder's Name: Zhang Baoshan/CH10 Identification: 340421198105094217 Method of Identification: ID Card Date and place of birth: 9 May 1981, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: ISO 9606-1:2017	Examiner or Test Body Reference No. iP1014-034-TK22-11-ZB 
Job Knowledge: Acceptable ☐ Not tested ☒	

	Weld test details	Range of approval
Welding process(es)	135	135
Transfer mode	globular	spray, globular
Product type (Plate or pipe)	Plate	Plate / Pipe
Type of weld	FW	FW
Parent Material group(s)	11.1 to 11.1	1~11
Filler Material group(s)	FM1 (S)	FM1, FM2 (S, M)
Shielding gases	75%Ar+25%CO ₂)	75%Ar+25%CO ₂
Auxiliaries (eg backing gases)	without	With or without
Type of current and polarity	DCEP	DCEP
Material thickness (mm)	6	≥3
Outside pipe diameter (mm)	--	≥75 (rotating), or ≥500 for fixed pipes
Welding position	PB	PA, PB
Welding details	ml	sl, ml

Supplementary fillet weld test is available on attached sheet and or document No. NA

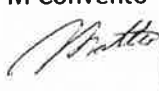
Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiographic testing	☐	☒
Fracture test	☐	☒
Bend test	☐	☒
Notch tensile test	☐	☒
Macroscopic exam	☒	☐
other	☐	☒

Examiner or test body: HPi Verification Services (Ireland)

Date of issue: 24 April 2022

Location: Co. Meath, Ireland

Name of Examiner: M Convento

Signature: 

Validity of approval until 18 Apr. 2025



Revalidation in accordance to

9.3 a) 3 YEARS

Prolongation for approval by examiner or test body for the following 2 years

Date	Signature	Position or title

Prolongation for approval by employer/co-ordinator the following 6 months

Date	Signature	Position or title
2022.9.28		welding engineer
2023.3.20		welding engineer
2023.9.15		welding engineer

	REFERENCE:	PRE - F - 24	ISSUE:	01
	TITLE:	ISO 9606-1 Welder Qualification Test Certificate		

Designation: 135 P Branch FM1 S t6 D48.3 PA+PB ss nb WPS No.: pWPS-TK22-07 Rev.0 Welder's Name: Zhang Baoshan/CH10 Identification: 340421198105094217 Method of Identification: ID Card Date and place of birth: 9 May 1981, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: ISO 9606-1:2017	Examiner or Test Body Reference No. iP1014-034-TK22-07-ZB 
Job Knowledge: Acceptable ☐ Not tested ☒	

	Weld test details	Range of approval
Welding process(es)	135	135
Transfer mode	globular	globular, spray
Product type (Plate or pipe)	Plate, Pipe	Plate/Pipe, Pipe
Type of weld	Branch (Set-through)	Branch
Parent Material group(s)	11.1 to 11.1	1~11
Filler Material group(s)	FM1 (S)	FM1, FM2 (S, M)
Shielding gases	75% Ar+25% CO2	75% Ar+25% CO2
Auxiliaries (eg backing gases)	Without	With or Without
Type of current and polarity	DCEP	DCEP
Deposited thickness (mm)	6	3 to 12
Outside pipe diameter (mm)	48.3	≥25
Welding position	PA+PB	PA, PB, PA+PB
Welding details	ss, nb	ss nb, ss mb, bs, ss gb, ss fb

Supplementary fillet weld test is available on attached sheet and or document No. NA

Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiographic testing	☐	☒
Fracture test	☐	☒
Bend test	☐	☒
Notch tensile test	☐	☒
Macroscopic exam	☒	☐
other	☐	☒

Examiner or test body: HPIVS Ireland
 Date of issue: 11 April 2022
 Location: Clonross, Ireland
 Name of Examiner: M Convento
 Signature: 
 Validity of approval until 05 April 2025



Revalidation in accordance to 9.3 a) 3 YEARS

Prolongation for approval by examiner or test body for the following 2 years

Date	Signature	Position or title

Prolongation for approval by employer/co-ordinator the following 6 months

Date	Signature	Position or title
2022.9.15		welding engineer
2023.3.20		welding engineer
2023.9.15		welding engineer

	REFERENCE:	PRE - F - 20	ISSUE:	03
	TITLE:	Approval Test Certificate for Welding Operator		

Designation: 121 P BW FM3 t12 PA Welding Procedure Specification Ref No.: pWPS-TK22-02 Rev.0 Qualification based on (see 4.1): PQR-TK22-02 Operator's Name: Qi Zhiguo/CH07 Identification: 340121198006183111 Method of Identification: ID Date and place of birth: 18 June 1980, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: EN ISO 14732:2013	Examiner or Test Body Ref. No. iP1014-034-TK22-02 
Job Knowledge: Acceptable ☐ Functional test: Acceptable ☒	

Weld test details		Range of approval
Welding process	121	121
Welding equipment	wire feeder	wire feeder
Welding unit	No	No
Type of weld	BW	BW
Details for mechanised welding:		
Visual control / visual remote control	Visual Control	Visual Control
Automatic arc length control	NA	NA
Automatic joint tracking	Without	With or without
Welding position	PA	PA
Single run/multi run	Multi run	Single run, multi run
Material backing	With	With
Consumable insert	No	No

Additional information is available on attached sheet and or weld procedure spec No. ---

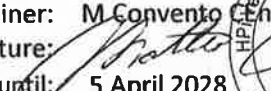
Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiography	☒	☐
Tensile test	☐	☐
Bend	☐	☐
Macro	☐	☐

Examiner or test body: HPIVS Ireland

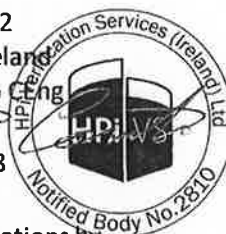
Date of issue: 11 April 2022

Location: Clonross, Ireland

Name of Examiner: M Convento CEng

Signature: 

Validity of approval until: 5 April 2028



Revalidation of qualification by examiner or examining body for [6 years (clause 5.3a)]

Confirmation of validity of qualifications by employer/co-ordinator for the following 6 months.

Date	Signature	Position or title

Date	Signature	Position or title
2022.9.28		welding engineer
2023.3.20		welding engineer
2023.9.18		welding engineer

	REFERENCE:	PRE - F - 24	ISSUE:	01
	TITLE:	ISO 9606-1 Welder Qualification Test Certificate		

Designation: 135 P BW FM1 S t6 PC ss nb WPS No.: pWPS-TK22-04 Rev.0 Welder's Name: Zhang Baoshan/CH10 Identification: 340421198105094217 Method of Identification: ID Card Date and place of birth: 9 May 1981, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: ISO 9606-1:2017	Examiner or Test Body Reference No. iP1014-034-TK22-04-ZB 
Job Knowledge: Acceptable ☐ Not tested ☒	

	Weld test details	Range of approval
Welding process(es)	135	135
Transfer mode	globular	globular, spray
Product type (Plate or pipe)	Plate	Plate, Pipe
Type of weld	BW	BW
Parent Material group(s)	11.1 to 11.1	1~11
Filler Material group(s)	FM1 (S)	FM1, FM2 (S, M)
Shielding gases	75%Ar+ 25%CO2	75%Ar+ 25%CO2
Auxiliaries (eg backing gases)	without	With or without
Type of current and polarity	DCEP	DCEP
Deposited thickness (mm)	6	3 to 12
Outside pipe diameter (mm)	--	≥75 (rotating), or ≥500 for fixed pipes
Welding position	PC	PA, PC, PA+PC
Welding details	ss, nb	ss, nb, ss, mb, bs, ss gb, ss fb

Supplementary fillet weld test is available on attached sheet and or document No. NA

Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiographic testing	☒	☐
Fracture test	☐	☒
Bend test	☒	☐
Notch tensile test	☐	☒
Macroscopic exam	☐	☒
other	☐	☒

Examiner or test body: HPiVS Ireland

Date of issue: 11 April 2022

Location: Clonross, Ireland

Name of Examiner: M Convento

Signature: 

Validity of approval until 05 April 2025



Revalidation in accordance to 9.3 a) 3 YEARS

Prolongation for approval by examiner or test body for the following 2 years

Date	Signature	Position or title

Prolongation for approval by employer/co-ordinator the following 6 months

Date	Signature	Position or title
2022.9.25		welding engineer
2023.3.20		welding engineer
2023.9.15		welding engineer

	REFERENCE:	PRE - F - 24	ISSUE:	01
	TITLE:	ISO 9606-1 Welder Qualification Test Certificate		

Designation: 135 P BW FM1 S t28 PC ss nb WPS No.: pWPS-TK22-10 Rev.0 Welder's Name: Zhang Baoshan/CH10 Identification: 340421198105094217 Method of Identification: ID Card Date and place of birth: 9 May 1981, China Employer: Well-Tanker (Anhui) M&E Equipment Co., Ltd. Code/ Test Standard: ISO 9606-1:2017	Examiner or Test Body Reference No. iP1014-034-TK22-10-ZB 
Job Knowledge: Acceptable ☐ Not tested ☒	

Weld test details		Range of approval
Welding process(es)	135	135
Transfer mode	globular	globular, spray
Product type (Plate or pipe)	Plate	Plate, Pipe
Type of weld	BW	BW
Parent Material group(s)	11.1 to 11.1	1~11
Filler Material group(s)	FM1 (S)	FM1, FM2 (S, M)
Shielding gases	75%Ar+25%CO2)	75%Ar+25%CO2
Auxiliaries (eg backing gases)	without	With or without
Type of current and polarity	DCEP	DCEP
Deposited thickness (mm)	28	≥3
Outside pipe diameter (mm)	--	≥75 (rotating), or ≥500 for fixed pipes
Welding position	PC	PA, PC, PA+PC
Welding details	ss, nb	ss, nb, ss, mb, bs, ss gb, ss fb

Supplementary fillet weld test is available on attached sheet and or document No. N/A

Type of Test	Performed and acceptable	Not required
Visual	☒	☐
Radiographic testing	☒	☐
Fracture test	☐	☒
Bend test	☒	☐
Notch tensile test	☐	☒
Macroscopic exam	☐	☒
other	☐	☒

Examiner or test body: HPiVS Ireland

Date of issue: 11 April 2022

Location: Clonross, Ireland

Name of Examiner: M Convento

Signature: 

Validity of approval until 05 April 2025

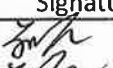


Revalidation in accordance to 9.3 a) 3 YEARS

Prolongation for approval by examiner or test body for the following 2 years

Date	Signature	Position or title

Prolongation for approval by employer/co-ordinator the following 6 months

Date	Signature	Position or title
2022.9.15		Welding Engineer
2023.3.20		Welding Engineer
2023.9.15		Welding Engineer